

DSA Day 37: Selection & Bubble Sort

Book: Data Structures Through C in Depth - Ch8 §8.1-8.9 · pp 417-427 | Code in Python

Overview

Selection sort repeatedly selects the minimum; bubble sort swaps adjacent out-of-order pairs. Both $O(n^2)$; bubble can stop early if sorted.

Key concepts to master

- stability, in-place
- best/worst/avg case

Python implementation

```
def bubble(a):
    for i in range(len(a)):
        swapped=False
        for j in range(len(a)-1-i):
            if a[j]>a[j+1]: a[j],a[j+1]=a[j+1],a[j]; swapped=True
        if not swapped: break
```

Complexity

$O(n^2)$; bubble $O(n)$ on already-sorted.

Common pitfalls

- Selection sort is NOT stable
- Forgetting the early-exit flag

Practice

- Sort {5,2,9,1,7}, count swaps
- Which is stable?

